



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

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Class : CSE
Batch : 2025-29

Assignment : 13

Subject: Programming with Java

ASSIGNMENT No : 13

TITLE : Write a Java program to perform reading from and writing to a file using file handling operations.

Problem Statement

Write a Java program to perform reading from and writing to a file using file handling operations.

Learning Objectives

To perform file reading and writing operations using Java file handling classes.

Theory

Java provides file handling through the java.io package. Classes such as File, FileReader, FileWriter, BufferedReader, and BufferedWriter are used to perform input and output operations. File handling enables permanent storage and retrieval of application data. Proper exception handling is necessary to manage file-related errors. These concepts are essential for data persistence.



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Code and Output Screenshots

```
J FileHandlingExample.java > Java > FileHandlingExample
1  import java.io.*;
2
3  public class FileHandlingExample {
4      Run main | Debug main | Run | Debug
5      public static void main(String[] args) {
6          try {
7              FileWriter fw = new FileWriter("data.txt");
7              fw.write("Hello, this is a Java file handling program.");
8              fw.close();
9
10             FileReader fr = new FileReader("data.txt");
11             BufferedReader br = new BufferedReader(fr);
12
13             String line;
14
15             while ((line = br.readLine()) != null) {
16                 System.out.println(line);
17             }
18
19             br.close();
20             fr.close();
21
22         } catch (IOException e) {
23             System.out.println("File error: " + e.getMessage());
24         }
25     }
26 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\LENOVO\Documents\JAVA Assignments> cd "c:\Users\LENOVO\Documents\JAVA Assignments\" ; if ($?) { javac FileHandlingExample.java } ; if ($?) { java FileHandlingExample }
Hello, this is a Java file handling program.
PS C:\Users\LENOVO\Documents\JAVA Assignments>
```

Data.txt :

```
data.txt
1  Hello, this is a Java file handling program.
```



Conclusion : Thus, the Java program successfully demonstrated file handling using FileWriter, FileReader, and BufferedReader. The program created a file, wrote data into it, and then read and displayed the contents of the file. It also handled file-related errors using exception handling.

Exercise

Q1) Create a program to write and read employee details from a file.

Ans) Code :

```
J A13E1.java > Java > A13E1
1  import java.io.*;
2
3  public class A13E1 {
4      Run main | Debug main | Run | Debug
5      public static void main (String[] args) {
6          try {
7              FileWriter fw = new FileWriter("employee.txt");
8
9              fw.write("Employee ID: 100\n");
10             fw.write("Name: Atharva\n");
11             fw.write("Salary: 85000\n");
12
13             fw.close();
14
15             FileReader fr = new FileReader("employee.txt");
16             BufferedReader br = new BufferedReader(fr);
17
18             String line;
19
20             while ((line = br.readLine()) != null) {
21                 System.out.println(line);
22             }
23
24             br.close();
25             fr.close();
26         } catch (IOException e) {
27             System.out.println("File error: " + e.getMessage());
28         }
29     }
30 }
```

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```
PS C:\Users\LENOVO\Documents\JAVA Assignments> cd "c:\Users\LENOVO\Documents\JAVA Assignments\" ; if ($?) { javac A13E1.java } ; if ($?) { java A13E1 }
Employee ID: 100
Name: Atharva
Salary: 85000
PS C:\Users\LENOVO\Documents\JAVA Assignments>
```



Employee.txt :

```
employee.txt
1   Employee ID: 100
2   Name: Atharva
3   Salary: 85000
4   |
```

Q2) Create a Java program to write patient details (patient ID, name, age, and diagnosis) to a file and read the details from the file for display.

Ans) Code :

```
J A13E2.java > Language Support for Java(TM) by Red Hat > A13E2
1   import java.io.*;
2
3   public class A13E2 {
4       Run main | Debug main | Run | Debug
5       public static void main (String[] args) {
6           try {
7               FileWriter fw = new FileWriter("details.txt");
8
9               fw.write("Patient ID: 200\n");
10              fw.write("Name: Manish\n");
11              fw.write("Age: 19\n");
12              fw.write("Diagnosis: Cough\n");
13
14              fw.close();
15
16              FileReader fr = new FileReader("details.txt");
17              BufferedReader br = new BufferedReader(fr);
18
19              String line;
20
21              while ((line = br.readLine()) != null) {
22                  System.out.println(line);
23              }
24
25              br.close();
26              fr.close();
27
28          } catch (IOException e) {
29              System.out.println("File error: " + e.getMessage());
30          }
31      }
32  }
```

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```
PS C:\Users\LENOVO\Documents\JAVA Assignments> cd "c:\Users\LENOVO\Documents\JAVA Assignments\" ; if ($?) { javac A13E2.java } ; if ($?) { java A13E2 }
Patient ID: 200
Name: Manish
Age: 19
Diagnosis: Cough
PS C:\Users\LENOVO\Documents\JAVA Assignments> |
```



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Details.txt :

≡ details.txt

```
1 Patient ID: 200
2 Name: Manish
3 Age: 19
4 Diagnosis: Cough
5 |
```